

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

Quiz no. 2

Summer Semester 2021-2022

DURATION: 30 MIN

FULL MARKS: 15

Math 4641: Numerical Methods

Figures in the right margin indicate marks.

1.

Suppose you are given the following data points:

x	0	0.5	1	1.5
$f(x)$	0	0.4794	0.8415	0.9975

Use the highest order polynomial possible for the given data points and find $f(0.6)$ using the Direct Interpolation Method.

7
(CO2)
(PO1)

2.

Find the Maclaurin Series (showing at least first 4 terms) for $\ln(1+x)$ and $\ln(1-x)$ and hence find the Maclaurin Series for $\ln\left(\frac{1+x}{1-x}\right)$.

8
(CO1)
(PO2)